

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q18-0937

Gresham 230 kV
April 23, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-0937. The Interconnection Customer proposes a 764.8 MW natural gas combined cycle facility interconnecting at the Gresham 230 kV substation in the Ohio area. The request was received on February 04, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 372 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$224,225,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
BRINCKEN'S CORNER - NARCOOSSEE 100 KV	NARCOOSSEE - LAWRENCE TAP 100 KV	155.4	142	17.0
NARCOOSSEE - PICNIC POINT 100 KV Cir. 1	NARCOOSSEE - PICNIC PT 100 KV Line 2	154.6	179	17.8
NARCOOSSEE - PICNIC POINT 100 KV	NARCOOSSEE - PICNIC PT 100 KV Cir. 1	154.6	179	17.8
FERN ST SUBSTATION 345/230kV Transformer Bank	PICNIC PT - GRESHAM 230 KV	151.1	438	100.0
CHILHOWEE - TERMINAL 100 KV #1	CHILHOWEE - TERMINAL 100 KV #2	148.1	158	9.8
PICNIC POINT - GRESHAM 230 KV Line	GRESHAM - AVON PARK NORTH 230 KV	136.5	523	100.0
MARLOW - BINGEN 138 KV	BAY POINT - ALGODONES 345 KV	131.9	234	5.9
TERMINAL - CHILHOWEE 100 KV	TERMINL - CHILHOWEE 100 KV Cir. 1	130.9	179	9.8
GRESHAM SUBSTATION - AVON PARK NORTH 230 KV	GRESHAM - PICNIC PT SUBSTATION 230 KV	127.7	558	100.0
HANGMAN - AVON PARK N 230 KV	PICNIC PT - GRESHAM 230 KV	120.2	573	100.0
FERN ST - HANGMAN 230 KV	PICNIC PT - GRESHAM 230 KV	115.8	571	100.0
PROGRESS PARK - CHILHOWEE 100 KV	OSAWATOMIE - SOUTHSIDE WASTEWATER TREATMENT PLANT 100 KV	115.7	155	5.9

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
SOUTSIDE WASTEWATER TREATMENT PLANT - HOT SPRINGS 100 KV line	BODO - BLUESTEM 230 KV	114.0	165	18.0
SOUTHSIDE WASTEWATER TREATMENT PLANT - OSAWATOMIE 100 KV	BLUESTEM - BODO 230 KV	112.6	192	14.7
STEVENS RD SUBSTATION - CONESVILLE STATION 230 KV line	CONESVILLE STA - MUSKINGUM RIV 500 KV	110.3	624	5.1
WAGNER LAKE - TYRANENA 138 KV	ALGODONES - BAY POINT 345 KV	108.1	213	5.4
NARCOOSSEE - LAWRENCE TAP 100 KV	AUTAUGAVILLE SUBSTATION - BRINCKEN'S CORNER 100 KV	106.8	142	17.4
CENTRAL - BINGEN 138 KV	OSAWATOME - WAGNER LAKE 230 KV line	106.7	373	10.8
CENTRAL - REDFORD 138 KV	OSAWATOMIE - WAGNER LAKE 230 KV	106.2	390	10.8
CHILHOWEE - OSAWATOMIE 100 KV Line	OSAWATOMIE - WAGNER LAKE 230 KV	105.6	140	13.4
REDFORD - NEW HOLLANDALE 138 KV	WAGNER LAKE - OSAWATOMIE 230 KV	105.6	411	10.8
TERMIAL - FINNEYTOWN 100 KV #1	TERMINAL - FINNEYTOWN 100 KV	104.7	153	5.4
TERMINAL - FINNEYTOWN 100 KV line	FINNEYTOWN - TERMINAL 100 KV	104.7	153	5.4
BOSWELL DAIRY AVE - FINNEYTOWN 100 KV	HORTONVILLE - PICNC POINT 100 KV	104.5	136	6.1
FARIBAULT - PROGRESS PARK 100 KV	SOUTHSIDE WASTEWATER TREATMENT PLANT - OSAWATOMIE 100 KV line	102.5	137	5.9

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
BRINCEN'S CORNER - NARCOOSSEE 100 KV	Rebuild line with higher-capacity conductor	\$1,860,000	41% (\$761,000)
PICNIC PT - NACOOSSEE 100 KV Cir. 1	Rebuild line with higher-capacity conductor	\$1,255,000	43% (\$541,000)
NARCOOSSEE - PICNIC POINT 100 KV #2	Rebuild line with higher-capacity conductor	\$1,180,000	43% (\$508,000)
FERN ST SUBSTATION 345/230kV Transformer Bank	Replace transformer with higher-rated unit	\$16,846,000	100% (\$16,846,000)
CHILHOWEE - TERMINAL 100 KV #1	Rebuild line with higher-capacity conductor	\$1,208,000	100% (\$1,208,000)
PICNIC POINT - GRESHAM 230 KV Line	Rebuild line with higher-capacity conductor	\$18,117,000	100% (\$18,117,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
MARLOW - BINGEN 138 KV	Rebuild line with higher-capacity conductor	\$6,118,000	19% (\$1,174,000)
TERMINAL - CHILHOWEE 100 KV	Rebuild line with higher-capacity conductor	\$1,670,000	100% (\$1,670,000)
GRESHAM SUBSTATION - AVON PARK NORTH 230 KV	Rebuild line with higher-capacity conductor	\$12,787,000	100% (\$12,787,000)
HANGMAN - AVON PARK NORTH 230 KV	Rebuild line with higher-capacity conductor	\$21,447,000	100% (\$21,447,000)
FERN ST - HANGMAN 230 KV	Rebuild line with higher-capacity conductor	\$134,969,000	100% (\$134,969,000)
PROGRESS PARK - CHILHOWEE 100 KV	Rebuild line with higher-capacity conductor	\$2,052,000	100% (\$2,052,000)
SOUTSIDE WASTEWATER TREATMENT PLANT - HOT SPRINGS 100 KV line	Rebuild line with higher-capacity conductor	\$1,303,000	30% (\$398,000)
SOUTHSIDE WASTEWATER TREATMENT PLANT - OSAWATOMIE 100 KV	Rebuild line with higher-capacity conductor	\$1,890,000	34% (\$635,000)
STEVENS RD SUBSTATION - CONESVILLE STATION 230 KV line	Rebuild line with higher-capacity conductor	\$47,806,000	11% (\$5,299,000)
TYRANENA - WAGNER LAKE 138 KV	Reconductor line	\$2,501,000	46% (\$1,164,000)
NARCOOSSEE - LAWRENCE TAP 100 KV	Reconductor line	\$646,000	36% (\$234,000)
CENTRAL - BINGEN 138 KV	Reconductor line	\$550,000	27% (\$151,000)
CENTRAL - REDFORD 138 KV	Reconductor line	\$550,000	27% (\$151,000)
OSAWATOMIE - CHILHOWEE 100 KV	Reconductor line	\$265,000	46% (\$121,000)
REDFORD - NEW HOLLANDALE 138 KV	Reconductor line	\$426,000	27% (\$117,000)
TERMINAL - FINNEYTOWN 100 KV	Reconductor line	\$1,176,000	100% (\$1,176,000)
TERMINAL - FINNEYTOWN 100 KV	Reconductor line	\$688,000	100% (\$688,000)
BOSWELL DAIRY AVE - FINNEYTOWN 100 KV	Reconductor line	\$1,021,000	100% (\$1,021,000)
FARIBAULT - PROGRESS PARK 100 KV line	Reconductor line	\$990,000	100% (\$990,000)
	Total allocated to project		\$224,225,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.